Amarthya Jonnalagadda

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Summary

Final-year CSE (AI & ML) student (Class of 2027) who builds, fine-tunes and **evaluates** AI systems: a live, solo-built multi-agent platform with 280+ tests and an eval harness, and a team-built Telugu voice assistant with a fine-tuned vision model (paper under review). Full-stack internship (Java, Spring Boot, React). Seeking full-time **AI/ML Engineer** or **Software Engineer** roles.

Education

VIT-AP University, B.Tech in Computer Science Engineering (AI & ML) 2023 – 2027

- **CGPA:** 8.55/10 | **Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Machine Learning

Dr. KKR's Happy Valley School, Class XII: 88% | **KKR Gowtham School**, Class X: 100% 2021 – 2023

Experience

Full Stack Web Developer Intern, GRAAS (Growth as a Service) June 2025 – July 2025

- Developed 2 full-stack web applications end-to-end using **Java, Spring Boot and React**, serving both internal and client-facing use cases
- Designed and implemented **10+ RESTful API endpoints** and integrated them with responsive React interfaces
- Debugged and resolved backend and frontend defects, reducing reported issues during QA cycles

Projects

Pathlight – Autonomous Career Intelligence Platform (Solo) Live | GitHub

- Built and deployed (Vercel, Render, MongoDB Atlas) an agentic platform that sources openings from job boards and placement emails, checks eligibility, finds skill gaps, tailors resumes, and applies **only after the user approves**
- Orchestrated **5 LangGraph agents** with **deterministic rules before any LLM call**, a bring-your-own-key LLM fallback chain (Gemini, Groq, OpenAI, Anthropic) and **4 MCP servers** (Gmail, GitHub, filesystem, reminders)
- Wrote an **evaluation harness** on hand-labeled benchmarks: discovery agent 100% field accuracy and 0.94 skills F1; eligibility agent 87.5% accuracy with **0 false positives**; used it to retune retrieval thresholds
- **287 automated tests** (222 pytest, 65 Vitest) with fake-LLM dependency injection; a code-level guard rejects any tailored resume that invents skills
- *Tech: Python, FastAPI, LangGraph, MCP, Gemini API, ChromaDB (RAG), MongoDB, Next.js, TypeScript, Docker*

Krishi-Agent – Telugu Voice Farm Advisory (Team of 4, Capstone) GitHub

- Voice assistant that answers farmers in spoken Telugu: speech recognition, a rules-first intent router, disease, weather and mandi-price agents, and a rule-based conflict resolver (paper under review)
- **Fine-tuned a DINOv2 ViT-L/14** crop-disease classifier on real field photos, improving accuracy from **23.5% to 72.9%** on PlantDoc field photos and **91.1%** on 2,640 unseen potato photos, with temperature-scaled confidence and refusal of unknown leaves
- Rules-first intent router hit **95.8%** vs 82.3% for the LLM alone (McNemar $p = 0.0002$); Telugu ASR at **11.7% WER**; 80/80 test questions answered without fallback
- *Tech: PyTorch, DINOv2, FastAPI, Ollama (Qwen2.5-7B), AI4Bharat ASR/TTS, data.gov.in API*

Quotation Management Web App (Real Client) GitHub

- Digitized product and quotation management for a family-owned home-theatre business with **Spring Boot, React and MongoDB**, replacing a manual Excel process with automated PDF quotations; deployed on Azure

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL

AI / LLM: LangGraph, Multi-Agent Systems, MCP, RAG (ChromaDB), Tool Calling, LLM Evaluation, Gemini API, Ollama

ML / Deep Learning: PyTorch, Fine-tuning Vision Transformers (DINOv2), Model Calibration, Statistical Testing (McNemar), Speech (ASR/TTS), Hugging Face

Backend & Frontend: FastAPI, Spring Boot, REST APIs, JWT Auth, React, Next.js

Data, Cloud & Testing: MongoDB, SQL, Docker, Azure, AWS, Vercel, Render, GitHub Actions, pytest, Vitest

Certifications: AWS Certified Cloud Practitioner, Scrimba Learn React